

Engineering Reboot 2026

Week-02 Day-06 Engineering Lab Manual v4.0

Conditionals

Governance Baseline: Engineering-Reboot-Governance-v4.1-Locked-Superseding

Curriculum Baseline: Week-02-Master-Workbook-v5.0-Locked-Final

Primary Deliverable: `decision_program.py`

Part 1 - Foundations

Mission Brief

Learn software decision-making through IF, IF-ELSE and IF-ELIF-ELSE structures.

Learning Objectives

- Understand conditions
- Understand True and False evaluation
- Create conditional programs
- Validate execution paths

Engineering Context

Conditionals appear in electrical systems, building automation, embedded systems and software engineering.

Repository Preparation

Verify repository state using git status.

Artifact Placement Matrix

Exercises -> Week-02\Exercises\Day-06\

Notes -> Week-02\Notes\Day-06\Notes.md

Journal -> Week-02\Journal\Day-06\Journal.md

Evidence -> Week-02\Day-06\Evidence\

Common Beginner Mistakes

- Using = instead of ==
- Incorrect indentation
- Improper if/elif/else ordering

Part 2 - Laboratory Exercises

Exercise 01 - high_temperature_alert.py

Uses input() to obtain temperature.

Displays Air conditioning recommended when temperature > 80.

Exercise 02 - temperature_status.py

Evaluates whether temperature is acceptable.

Exercise 03 - age_classification.py

Classifies user as Minor or Adult.

Exercise 04 - grade_evaluation.py

Assigns Grade A, B, C or Needs Improvement.

Exercise 05 - decision_program.py

Evaluates voltage as:

Voltage Low

Voltage Normal

Voltage High

All exercises use:

NUMERIC INPUT

input()

int()

Evidence files are required for each exercise.

Part 3 - Documentation, Git Workflow and Audits

Notes.md

Location:

Week-02\Notes\Day-06\Notes.md

Journal.md

Location:

Week-02\Journal\Day-06\Journal.md

Validation Matrix

Temperature, Age, Grade and Voltage validation cases.

Evidence Matrix

All code and output screenshots.

Git Workflow

git status

git add .

git commit -m "Day 06 conditionals completed"

git push

Repository End State

Exercises, Notes, Journal and Evidence folders verified.

Completion Audit

Deliverables

Documentation

Evidence

Git Verification

GitHub Verification

Day 07 Preview

Boolean Logic

AND

OR

NOT